0.1 (U) Sample Component Section Overview

TIMELINE: Should be filled in prior to Preliminary Design Review (PDR) with final edits by Critical Design Review (CDR).
(U) In the context of this document, component refers to a functional block of code that is available within the software model that is representing a real-world capability or effect. In most cases, component refers to one of the blocks in the above functional diagram which is contained in the User and Analyst Manual (UAM) document and/or Unified Modeling Language (UML) diagram contained within the Architecture Description Document (ADD). Note for most cases, if there is a block in the above UML or block diagram, there should be a component (or multiple) component sections for each block.
(U) Describe the purpose of component. This section will describe what the component is doing from a high level, and it will more than likely include a graphic describing the nature of the component.
(U) For example, if the component is focused on the antenna. This may describe that it is a phased array antenna with 100 elements that is scanning in azimuth and elevation electrically and rotating mechanically at 6 rotations per minute. Be clear which pieces are the real system and which pieces are included within the model component.
(U) In the above UML example, there is a jammer processor which should be discussed as a component, and then individual components for each jamming mode (barrage, spot, etc.)

0.1.1 (U) Sample Component Section Functional Behavior

TIMELINE: Should be filled in prior to PDR with final edits by CDR.
(U) This section is focused on the functional side of the component and what it is representing in the RF domain as well as any math or data behind the component. This section is expected to be more focused on the physics and engineering behind what is going on with the component.
(U) Make sure to include the analytic background of the component to include detailing the algorithmic, mechanical/hardware, operational, or decision-making behavior(s) of the component. Examples of each of these include the following ideas:
o Algorithmic behavior can be considered as a behavior which can be described mathematically: Dwell Time Calculation, Signal Prioritization
o Mechanical/Hardware behavior can be considered as a behavior defined through physical properties: Constant rotation/translation rate, Constant gain presented, Constant bandwidth limitations, Maximum rotation/translation rate/distance
o Operation behavior can be considered as a behavior defined by a sequential series of events: radar signal processing sequences
o Decision-making behavior can be considered as a behavior defined by branching paths based on a condition: Dynamic mode switching, Dynamic signal adjustment
(U) For example, if the component is focused on the antenna, this section may focus that the phased array is represented as a sum pattern of the elements and to capture the electrical steering is just pointing the sum pattern in that direction and applying a roll-off to the gain instead of steering it. It would include antenna patterns, measured beamwidths, and comparisons to measured data.
(U) Please include high-level picture demonstrating the purpose of this component as a reference to walk through the code as necessary.

0.1.2 (U) Sample Component Section Code Structure

TIMELINE: Should be finalized by CDR.
(U) This section is focused on the code and use of the component within the model. Key information is which function(s) are associated with this component and data inputs and outputs that are coming from this component.
(U) This section should describe all public interfaces to the component along with necessary pre-conditions of things external to the component and arguments passed to those interfaces for meaningful results. The behavior of the interface should be detailed enough to enable a developer utilizing those interfaces to understand the expected outcome of a proper use of the interface such as changes in state of the component, modifications to arguments passed by reference or pointer to the interface, or return values.
(U) Include algorithms (pseudocode), UAM sequence diagrams, statechart diagrams etc. as applicable to convey the information (i.e., it may not be necessary to document algorithms such as accessors, mutators, and other simple algorithms). This section is focused on the computer science of the component details.
(U) For example, if the component is focused on the antenna, this section may focus that the phased array is represented by using the 3D antenna lookup code that is native to the architecture and include specific details as requested above.
(U) Additional examples include:
1) If a component has a reference to the spdw_inbox that is not passed as part of an interface function and assumes that there are Simulated Pulse Descriptor Word (SPDW) there, this would be a precondition.
2) If a list of items is passed by reference to an interface of the component which manipulates that list, that should be captured so the caller of the interface is aware.
(U) Do NOT include code snippets as it can change over time and currently trying to keep overhead document costs down.

0.1.3 (U) Sample Component Section Assumptions, Limitations, and Gaps

TIMELINE: Should be filled in prior to PDR with final edits by CDR.
(U) Capture any assumptions, limitations, gaps, or design decisions around this component that may impact analysis use or future re-use.
(U) For the antenna example, this can include that due to how the steering is captured, that the broadening of the mainbeam and any sidelobe impacts will not be captured in the process. This can impact Jamming-to-Signal Ratio (J/S) calculations or impacts of sidelobe jammers.

********** EXAMPLE COMPONENT 1 **********

0.2 (U) Early Warning Component Overview

THIS IS INCLUDED IN BOTH UAM AND ADD.

(U) At a high level, this component of the Overarching Dynamic Electronic-Warfare System Standard Architecture (ODESSA) model connects the functionality for the Early Warning Receiver (EWR) path to include the pedestal, antenna, RIF, angle measurement, Doppler filtering, and detection queue to send to the ModelNameGoesHere Computer component.

(U) Specifically, EWR path of the ModelNameGoesHere is a passive receiver focused on detection of emissions within the operational frequency band of the system from 8-12GHz . The EWR is rotating at 50rpm in order to quickly detect fast-moving targets such as missiles. While no targets are actively being jammed, the system

is continuously listening for emissions.

(U) If an emission is detected, it will be passed to the jammer to detect and automatically jam through a separate receiver/transmitter, and the EWR will continue to rotate and listen. While the jammer is emitting, the EWR will be blanked and unable to receive any emissions to avoid destroying components. See limitations below.

(U) At a high level, this component connects the functionality for the EWR path to include the pedestal, antenna, RIF, angle measurement, Doppler filtering, and detection queue to send to the Computer component.

0.2.1 (U) Early Warning Component Functional Behavior

THIS IS INCLUDED IN ONLY UAM.

(U) Attempts were made to make the EWR path as configurable as possible to enable future intel updates to further refine. Many inputs will be driven through a "Super User" input capability with a CSV file. Those parameters are contained below:

Table 1: (U) EWR Inputs

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Input Name	Default
EwrBandwidthHz	4GHz
EwrCenterFrequencyHz	10GHz
EwrNumberDopplerBins	32
SnrThresholdDb	13dB
NoiseFigureDb	6dB
ReceiverNoiseFigureDb	6dB
TransmitLossDb	2dB
EwrPeakGainDbi	10dbi
MonopulseAzimuthMeasurement	1.5deg
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(U) There is limited information on the processing time of the system, but we have a rotation rate and an expected azimuth beamwidth of the system. Since the system is not scanning in elevation, we can treat each azimuth position as a discrete dwell processing time for the system. The system does have monopulse processing capability, so it is expected the system will listen at smaller angle increments than the beamwidth resolution of 5deg . We are assuming that the angle step will be 30 percent of the total beamwidth giving a step size of 1.5deg .

(U) Using this information a dwell time can be calculated using the following equations:

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(U) Determine Azimuth Steps

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$$azimuthSteps = 360/1.5deg \quad (1)$$

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(U) Determine Rotation Per Second

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$$rotationTimeS = 60/50rpm \quad (2)$$

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(U) Determine Dwell Time

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$$dwellTime = rotationTimeS/azimuthSteps = 5ms \quad (3)$$

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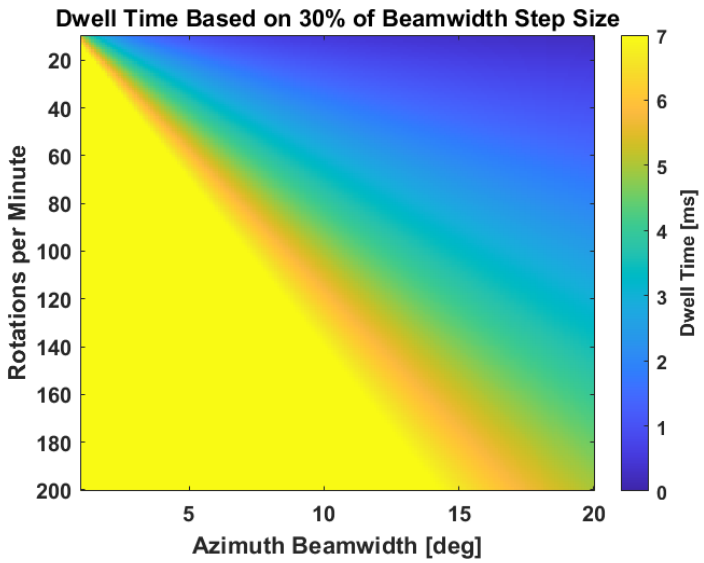


Figure 1: (U) EWR Dwell Time

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(U) Figure 1 demonstrates the dwell time impacts based on the beamwidth, 30% beamwidth step size, and rotation rate. For 5deg , step size of 1.5deg , and 50rpm , the dwell time is 5ms for the system. In order to support integration with future models which update at a slower rate than the dwell time, as well as concerns for meeting the real time requirement for the model, the decision was made to process all of the potential emissions within a full 360 degree rotation, and then time tag the SPDW's with the appropriate received time as if the EWR had rotated.

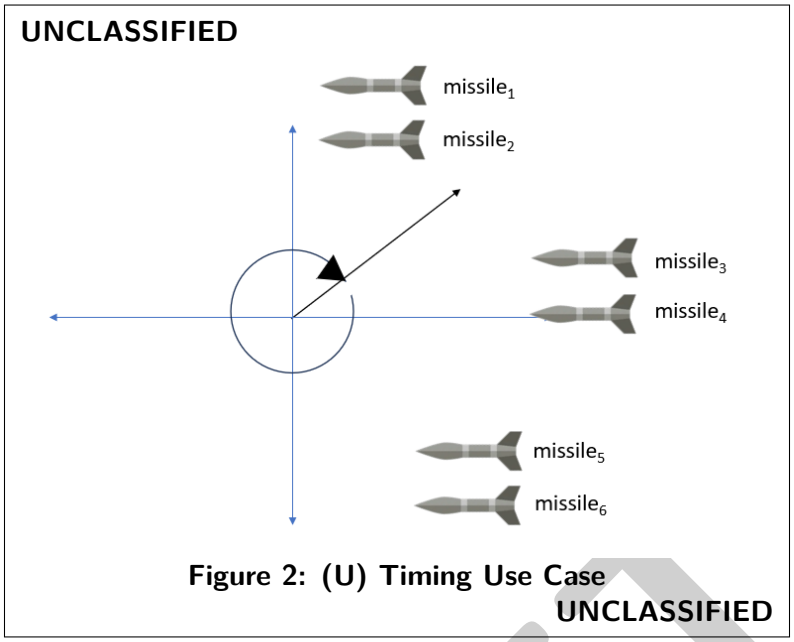


Figure 2: (U) Timing Use Case

(U) Using the figure shown in Figure 2, we can step through an example where the EWR is starting at an initial pointing angle, and there are six missiles in the scenario. In this example, we are assuming all missiles are within the operating band of the system and the signals are sufficient to be detected. All six of the SPDW will be processed, and then tagged with the appropriate time tag shown below in Figure 3.

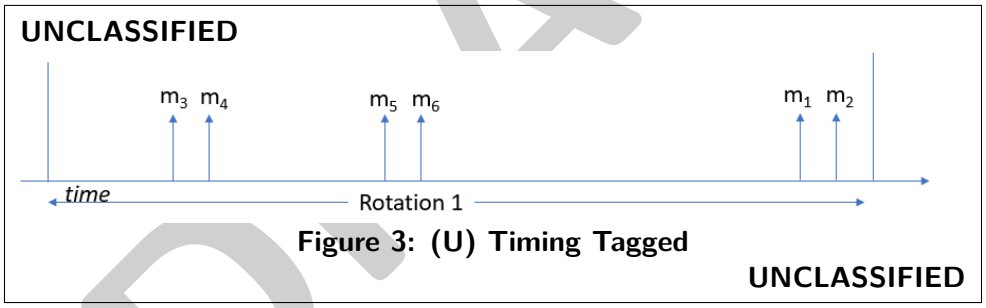


Figure 3: (U) Timing Tagged

(U) Using the Previous example, if the jammer is cued to missile 3 to start jamming, then the EWR will be blanked to receiving signal which is shown visually in Figure 4 where the blue boxes represent where the EWR was blanked.

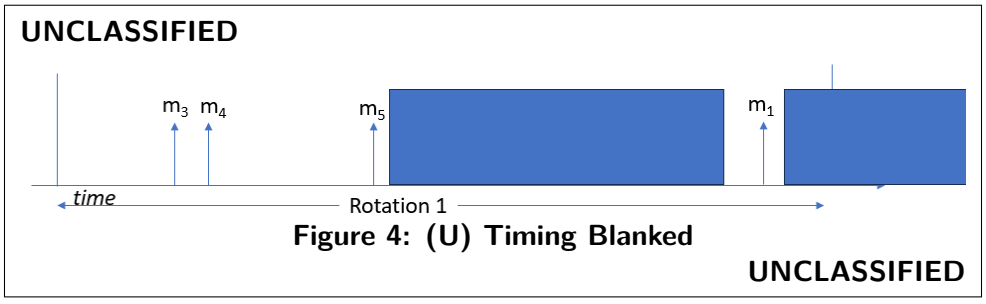


Figure 4: (U) Timing Blanked

In this case, the emissions from Missile 6 and Missile 2 are missed while the EWR is blanked.

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0.2.2 (U) Early Warning Component Code Structure

THIS IS INCLUDED IN ONLY ADD.

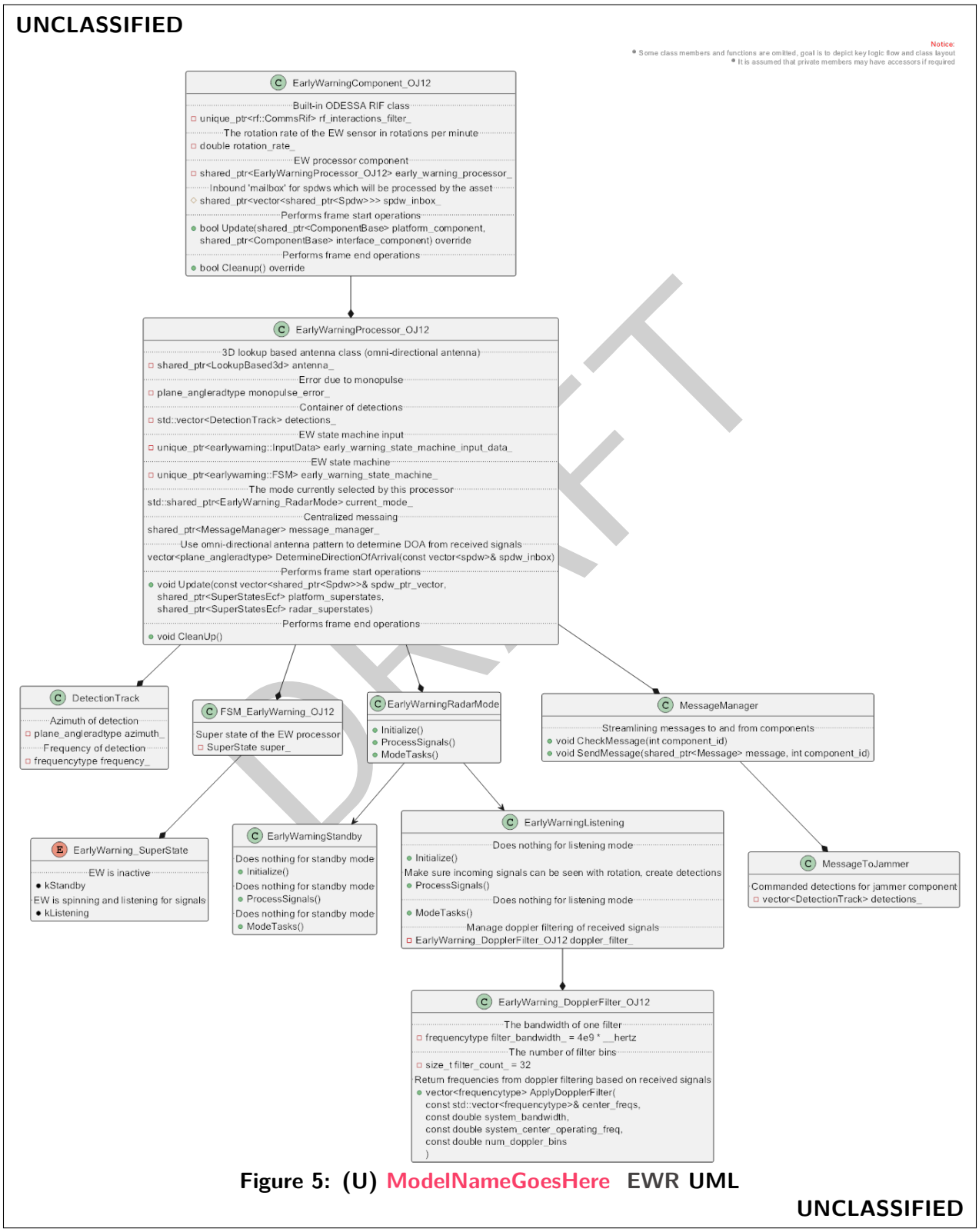


Figure 5: (U) ModelNameGoesHere EWR UML

(U) The EWR component contains the geometry conditions for the vehicle and records those states, initializes the modes (standby mode and listening mode), and creates the EWR processor.

(U) The processor manages the timing and modes of the system to account for the antenna psuedo-rotation process (more details below) and whether the system is in listening mode (EarlyWarningListening) or blanked during jamming (EarlyWarningStandby).

(U) The EarlyWarningListening.cpp contains the functions for managing the spdw_inbox for detections, managing the timing to verify the beam would be pointed at the entity when the signal arrives, processing signals through the RF Interactions Filter (RIF), calculating the angle measurement error, and applying the Doppler filter measurement.

(U) The EarlyWarningStandby.cpp is used when the EWR is blanked during jammer emission periods. While the EWR continues to "spin," no signals will be processed during the blanking period.

0.2.3 (U) Early Warning Component Unique Design Characteristics and Decisions

THIS IS INCLUDED IN ONLY UAM and is now "Assumptions, Limitations, and Gaps".

(U) It was decided to configure the code to have a single entity with a single entity controller, since the EWR and jammer are on a single truck. Then having two components will enable the two receiver paths and represent the EWR and jammer functions. This was based on discussions with the ODESSA architecture lead.

(U) In order to enable processing a full rotation, the antenna will be represented as an psuedo-omni antenna pattern. This will be discussed more in the antenna section.

(U) While the EWR is blanked, it may result in missed emissions in other angles for the EWR, and this will be captured after the signals have been processed through the RIF in the code.

********** EXAMPLE COMPONENT 2 **********

0.3 (U) EWR Antenna Overview

THIS IS INCLUDED IN BOTH UAM AND ADD.

(U) The ModelNameGoesHere EWR has a Phased Array that provides an azimuth beamwidth of 5deg and elevation beamwidth of 45deg .

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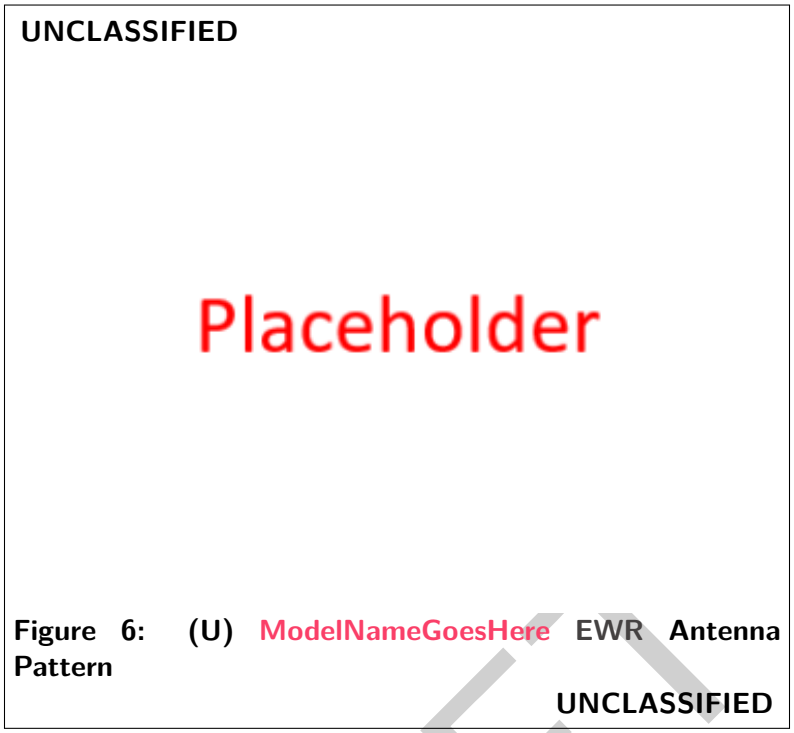


Figure 6: (U) ModelNameGoesHere EWR Antenna Pattern

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0.3.1 (U) EWR Antenna Functional Behavior

THIS IS INCLUDED IN ONLY UAM.

(U) In order to support the decisions outlined in the Early Warning Component Overview section, the antenna is being captured as a psuedo-omni pattern shown in Figure 6. This allows the antenna pattern to capture roll off in the elevation limiting it to the 45deg . However, all azimuth angles will have the peak gain of 10dbi .

(U) The ModelNameGoesHere has multiple elements which allow for monopulse processing in the azimuth plane to provide more refined measurements using monopulse processing. More details are contained within the monopulse error section on how this is handled.

(U) Antenna patterns were generated every gigahertz across the 8-12GHz range of the ModelNameGoesHere model. The antenna patterns are normalized to the CSV super user input, JammerPeakGainDbi, allowing users the ability to adjust the capabilities of the ODESSA jammer model. Between this aspect and having a pattern every gigahertz, we can capture the realistic variability of peak gain while providing a user with the power to experiment with model performance.

0.3.2 (U) EWR Antenna Code Structure

THIS IS INCLUDED IN ONLY ADD.

(U) The antenna is using the antenna lookup 3d base class and is called within the ModelNameGoesHere code within the EarlyWarning_Listening function. Antenna patterns are normalized to the CSV super user input, JammerPeakGainDbi, allowing users the ability to adjust the capabilities of the ODESSA jammer model.

(U) Using SPDWs that have passed the RIF, the model will first apply Doppler filtering to the pulses and determine a frequency bin. The model will then capture the receive time (spdw_received_pulse_start) and end time of the pulse (spdw_received_pulse_end) as well as the true direction of the emission using GetSpdwDirectionOfArrival. Then a dwell angle is calculated based on the current time using the GetDwellAzimuthCenter function. This will also take into account the pointing angle of the system itself to use as the starting angle of the EWR rotation (often listed as initial_angle within the source code). A boresight error calculation will be performed with the CalculateMonopulseBSE function within the EarlyWarningListening class and this error will be applied to the truth direction of arrival information. Then the direction of arrival information with boresight error is passed to the IsPulseIntercepted function to verify that the antenna would be pointed in the direction of the target when the emission was received. If so, then the detection is added to the track box. This is performed for all SPDWs that have been received within a full 360° scan.

0.3.3 (U) EWR Antenna Unique Design Characteristics and Decisions

THIS IS INCLUDED IN ONLY UAM and is now "Assumptions, Limitations, and Gaps

(U) Decisions outlined in Early Warning Component Overview drove representing the antenna pattern this way. This means that emissions in the sidelobe, while the EWR is pointed at another target, will not interfere with the mainbeam received signal. Each signal will be received and processed independently.

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1 (U) Acronyms

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Term	Description	Page
ADD	*Architecture Description Document*	1
CDR	*Critical Design Review*	1, 2
EWR	*Early Warning Receiver*	2, 3, 4, 5, 6, 7, 8, 1
J/S	*Jamming-to-Signal Ratio*	2
ODESSA	*Overarching Dynamic Electronic-Warfare System Standard Architecture*	2, 7, 8
PDR	*Preliminary Design Review*	1, 2
RIF	*RF Interactions Filter*	7, 1
SPDW	*Simulated Pulse Descriptor Word*	2, 5, 1
UAM	*User and Analyst Manual*	1, 2
UML	*Unified Modeling Language*	1

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